

ULTRASOUND MODES, FREQUENCY SELECTION, CALIBRATION, QUALITY INDICES, AND ARTIFACT REJECTION FOR RELIABLE INTERNAL ENDPOINT DISTANCE (NON-LIMITING)

A. Ultrasound Operating Modes (Non-Limiting)

In various embodiments, Node 4 uses one or more ultrasound operating modes to detect an internal endpoint and estimate distance. Non-limiting modes include: 1. Pulse-echo A-mode distance estimation along a longitudinal axis; 2. Multi-pulse ranging where repeated pulses are transmitted and averaged to improve stability; 3. Multi-angle interrogation using more than one transducer orientation to reduce specular reflection sensitivity; and 4. Power/range stepped operation, wherein transmit amplitude and/or pulse repetition rate is adjusted to improve detection across varying coupling conditions. In certain embodiments, the system uses at least one A-mode channel and produces a distance metric and a coupling/quality metric per window.

B. Frequency Bands and Tradeoffs (Non-Limiting)

In various embodiments, ultrasound frequency is selected based on target range, resolution needs, and coupling considerations. Non-limiting frequency selections include: 1. lower MHz-range operation for improved penetration and robust endpoint detection; 2. higher MHz-range operation for improved axial resolution at shorter distances; and 3. multi-frequency operation where the device cycles frequencies and selects the most stable endpoint response. In certain embodiments, the system uses frequency-dependent confidence scoring and selects outputs from frequency bands meeting stability thresholds.

C. Range, Timing, and Speed-of-Sound Considerations (Non-Limiting)

In various embodiments, Node 4 estimates distance using pulse timing and an assumed or adjusted speed of sound. Non-limiting approaches include: 1. using a nominal speed-of-sound value for soft tissue media; 2. applying a correction factor based on measured temperature context; 3. using calibration routines to map time delay to distance indices; and 4. reporting distance as a range band and confidence indicator rather than an absolute length when uncertainty is elevated. In certain embodiments, the system outputs both a primary distance estimate and an uncertainty/confidence estimate.

D. Calibration and Self-Check (Non-Limiting)

In various embodiments, Node 4 performs calibration and self-check routines to improve repeatability. Non-limiting routines include: 1. verifying transducer function via loopback checks and/or reference echo checks; 2. establishing baseline coupling signatures during a guided initialization step; 3. storing calibration coefficients for time-to-distance mapping; and 4. periodic drift checks using internal references and/or repeated short-range echoes. In certain embodiments, calibration is performed without requiring imaging output and is presented to the user as an abstract quality confirmation.

E. Coupling Quality Indices (Non-Limiting)

In various embodiments, Node 4 computes coupling quality indices from ultrasound echo characteristics. Non-limiting indices include: 1. echo amplitude stability across repeated pulses; 2. signal-to-noise ratio (SNR) estimates; 3. echo envelope consistency and peak sharpness; 4. dropout rate and saturation detection; and 5. continuity of endpoint estimates across sequential windows. In certain embodiments, a

coupling quality index is used to suppress or down-weight endpoint distance outputs when coupling is insufficient.

F. Artifact Rejection and Motion Gating (Non-Limiting)

In various embodiments, Node 4 uses motion and plausibility gating to reduce spurious distance estimates. Non-limiting methods include: 1. IMU-based gating: suppressing measurements during rapid motion, high jerk, or orientation instability; 2. rejecting endpoint estimates that violate continuity constraints across time; 3. rejecting echoes that are inconsistent with allowable range bounds; and 4. using redundant sensing (e.g., capacitive contact confirmation or circumferential contact sensing) to confirm the device is in contact and positioned as expected. In certain embodiments, the system generates a per-window validity label (valid / uncertain / invalid) and a confidence indicator.

G. Multi-Reflection Handling and Endpoint Selection Logic (Non-Limiting)

In various embodiments, the ultrasound channel may receive multiple echoes from internal surfaces. Non-limiting endpoint selection strategies include: 1. selecting the earliest strong reflector beyond a minimum distance; 2. selecting a stable peak that persists across multiple pulses; 3. using peak clustering or hysteresis to avoid rapid switching between reflectors; and 4. applying shape/consistency heuristics to reject multipath artifacts. In certain embodiments, the system reports an endpoint distance only when endpoint selection remains stable across a minimum number of pulses.

H. Integration with Controlled Radial Stimulus (Optional, Non-Limiting)

In optional embodiments, ultrasound endpoint sensing is combined with controlled radial stimulus (e.g., inflatable cuff/microbladder). The system may: 1. measure how endpoint distance changes under controlled expansion; 2. compute compliance indices from pressure vs. displacement relationships; and 3. use ultrasound-derived stability as a confirmatory signal for stimulus-response characterization.

I. Privacy-Preserving Outputs and Reporting (Non-Limiting)

In various embodiments, ultrasound outputs are presented as abstracted metrics (distance index, stability index, coupling quality index, confidence indicator) rather than anatomical images. Raw echo signals may be processed transiently and discarded after feature extraction unless explicitly enabled.

J. Variations

Any embodiment described herein may vary frequency, pulse timing, transducer configuration, and processing while maintaining the functional intent of reliable ultrasound endpoint distance estimation with coupling and motion gating.